

FLOOD PREDICTION SYSTEM

A Machine Learning Based Web Application
for Flood Risk Prediction

Submitted as part of APSICHE AI/ML Internship

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Technologies Used	Python, Flask, Machine Learning, HTML, CSS
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2. Certificate

This is to certify that the project titled “**Rising Waters – Flood Prediction System**” is a bonafide work carried out by **Anil Kumar Palla** as part of the **APSCHE AI/ML Internship**.

3. Acknowledgement

I would like to express my sincere gratitude to APSCHE and SmartBridge for providing this valuable opportunity to work on this project.

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Finally, I extend my thanks to all open-source contributors and Kaggle for making the dataset available for research and academic use.

4. Abstract

Floods are one of the most destructive natural disasters, causing significant loss of life and property every year. Early and accurate prediction of flood events is essential for effective disaster management and timely evacuation planning.

This project presents a Machine Learning-based Flood Prediction System that analyzes weather and rainfall data to predict flood risk. A Random Forest Classifier is used to train the predictive model, and a Flask-based web application is developed to provide real-time, user-friendly flood risk predictions.

5. Introduction

Flood prediction plays a critical role in modern disaster management systems. Traditional forecasting methods rely heavily on manual analysis and periodic meteorological reports, which can be slow and prone to human error.

With advancements in Artificial Intelligence and Machine Learning, predictive models can now analyze large volumes of historical data to generate fast and accurate predictions.

This project aims to build an AI-powered flood prediction system using historical rainfall data, enabling faster and more reliable flood risk assessment.

6. Problem Statement

Floods occur frequently due to heavy rainfall and changing climate patterns. Existing forecasting systems often lack the real-time prediction accuracy required for effective early warning.

There is a growing need for:

- Automated, data-driven flood prediction systems
- Faster decision-making tools for disaster response teams
- AI-based early warning systems accessible to the public

7. Objectives

- To predict flood occurrence using Machine Learning algorithms
- To build a user-friendly web interface for real-time predictions
- To analyze historical rainfall and weather patterns
- To improve the effectiveness of early warning systems
- To deploy the trained ML model using a Flask web application

8. Literature Survey

A review of existing approaches to flood and disaster prediction was conducted, covering the following areas:

- Traditional meteorological forecasting systems and their limitations
- Statistical rainfall analysis methods used in early warning systems
- Applications of Machine Learning models in natural disaster prediction
- Use of Random Forest algorithms in classification problems similar to flood risk assessment

9. Technology Stack

Category	Technology
Programming Language	Python
Web Framework	Flask
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
Model Serialization	Joblib
Frontend	HTML, CSS
Development Environment	Jupyter Notebook

10. Dataset Description

Source: Kaggle Rainfall Dataset

Features Used

- Temperature
- Humidity
- Cloud Cover
- Seasonal Rainfall (Jan–Feb, Mar–May, Jun–Sep, Oct–Dec)
- Annual Rainfall
- Average Rainfall

Target Variable

- Flood (0 = No Flood, 1 = Flood)

11. System Architecture



Workflow

- User enters weather data through the web interface
- Flask application sends the input data to the trained model
- The Machine Learning model predicts the flood risk
- The prediction result is displayed on the web page

12. Methodology

- Data Collection
- Data Preprocessing
- Feature Selection
- Model Training (Random Forest)
- Model Evaluation
- Web Deployment using Flask

13. Algorithm Used

Random Forest Classifier

Random Forest is an ensemble learning method that builds multiple decision trees during training and combines their outputs to produce a more accurate and stable prediction.

Advantages

- High accuracy
- Handles large datasets efficiently
- Reduces overfitting

14. Implementation

Backend

- Python Flask handles incoming web requests
- The trained model is loaded using Joblib

Frontend

- An HTML form collects user input
- CSS is used to improve the overall UI design

Model Flow

- Input is received from the web form
- Input is converted into an array format
- The array is passed to the trained model
- The model returns the prediction result

15. Results

- The model successfully predicts flood risk based on input weather data
- The web interface provides real-time prediction output
- The system performs with good overall accuracy

Sample Outputs

Input Condition	Predicted Output
Normal rainfall levels	■ No Flood
Heavy / abnormal rainfall levels	■■ Flood Likely

16. Advantages

- Fast and efficient prediction
- Easy to use web interface
- AI-based intelligent system
- Helps in disaster preparedness and planning

17. Limitations

- Depends heavily on the quality of historical data
- No real-time weather API integration in the current version
- Prediction accuracy is dependent on dataset quality

18. Future Scope

- Integration with real-time weather APIs
- Development of a mobile application
- Cloud deployment for wider accessibility
- Use of advanced deep learning models for improved accuracy
- GIS-based flood mapping for visual risk representation

19. Conclusion

The Flood Prediction System demonstrates how Machine Learning can be effectively used for natural disaster prediction. The system provides quick and efficient flood risk analysis and lays a strong foundation that can be further improved for real-world deployment.

With future enhancements such as real-time weather data integration and advanced modeling techniques, this system has the potential to become a valuable tool in flood disaster management and early warning.